

# Multi-Nodal Energy Systems Modelling Scenarios of South Africa's Power Sector

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**Abstract**— Climate change has brought about a revolution in power sectors worldwide with an urge to increase renewable energy penetration in the generation mix. South Africa, which has conventionally been a coal-dependent power generation country, is transitioning to cleaner resources such as solar and wind, which are abundantly available in the country. This large-scale integration of renewable energy in the national grid brings about the need to increase grid flexibility and strengthening to ensure that the country's power demands are adequately met. This paper thus models South Africa's energy mix in the year 2031 according to their respective supply areas as mentioned in Eskom's Integrated Resource Plan (IRP) 2019 and Generation Connection Capacity Assessment (GCCA) 2023 report, using IRENA's FlexTool software and assesses the transmission bottlenecks that the system might encounter. The results indicate a higher share of renewable energy in the future system, improvements in the unmet demand, reduction of carbon dioxide emissions and reductions in renewable energy curtailment with developments in future plant allocations, battery storage capabilities and transmission line expansions (i.e., investing further in upstream network strengthening to harness a higher yield of renewable generation capacity from the greater Cape areas of South Africa).

**Keywords**— Energy systems modelling, FlexTool, load shedding, power sector, renewable energy, South Africa

## I. INTRODUCTION

Climate change is becoming a huge threat to humanity and a global emergency [1]. We can already see some of its adverse effects with increasing climate uncertainties, heat waves, draughts, floods and many other such events. These have triggered countries worldwide to pledge to reduce their greenhouse gas (GHG) emissions which are a major contributor to global warming by decarbonizing their major polluting sectors [2].

In South Africa, the power sector is the key contributor to the overall emissions in the country, accounting for around 40% (~225 MtCO<sub>2</sub>e) of overall emissions [3]. This is because South Africa is and has been a coal-dependent power generation country. Fig. 1 [4] indicates this generation plant mix for the year 2021 under ESKOM (South Africa's public electricity utility which supplies approximately 90% of South

Africa's electricity and generates roughly 30% of the electricity used in Africa). Fig. 1 shows that around 85% of the country's installed capacity under ESKOM in 2021 was derived from thermal coal fired power plants.

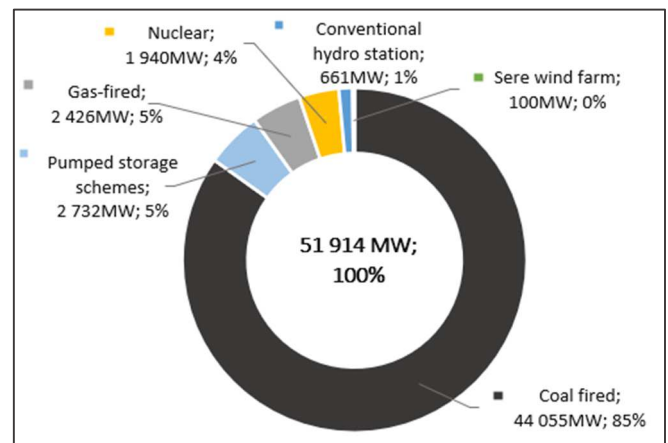


Fig. 1. Eskom's installed generation mix in 2021 (nameplate capacity in MW shown here) [4]

The transition to cleaner, renewable resources will be crucial for the decarbonization of South Africa's power sector. South Africa is abundantly rich in solar and wind resources which can be harnessed to produce electricity in the country. With recent and future improvements in renewable energy penetration and storage capacities, it is likely that the renewable share in the generation mix for South Africa's future power system would increase. This paper thus models South Africa's power system for the year 2031, which is mainly derived from the Integrated Resource Plan (IRP) 2019 [5] and the Transmission Development Plan (TDP) 2022-2031 [6], and analyzes the flexibility and transmission bottlenecks the system might encounter.

## II. ENERGY SYSTEMS MODELLING

### A. Overview

Energy systems modelling guides the operation of energy systems in conjunction with sector-specific models to provide insights into complex challenges such as climate mitigation

responses, ensuring energy security and maintaining grid flexibility [7]. Energy systems modelling of the power sector has been widely used across the globe to assess, simulate, and plan for a future power sector and how it would perform under various energy mix scenarios and different transition pathways [8]–[11].

The study in [9] modelled a 28-node European Union (EU) power network representing 28 EU member states. The PLEXOS framework was used to investigate challenges in the 2030 European power system as the EU transitions towards a low carbon energy system. The study in [10] modelled a 5-node multi-regional model of the Indian power system geared towards energy forecasts of year 2040 to evaluate six different scenarios to assess the effects of battery storage and renewable integration on the Indian power system using the Calliope model framework. The study in [11] modelled a 10-node multi-regional optimization planning model of China’s power sector using the linear programming solver of the General Algebraic Modelling System (GAMS) to optimize China’s future power sector in meeting demand whilst considering the impact of GHG emissions and carbon mitigation policy scenario. The energy modelling approach in this paper was adapted from [9,10, and 11], based on a widely accepted methodology and used IRENA’s FlexTool modelling tool for the implementation, modelling and analyses of South Africa’s power sector for the year 2031.

### B. Coal-fired plants in South Africa

South Africa’s power sector is largely coal-based and has a large fleet of fifteen coal-fired power plants which are mainly located in Mpumalanga. However, a lot of these plants are ageing and are scheduled for decommissioning in the next decade. Table I [12] shows a list of the nominal capacities (also known as sent-out capacity which refers to the difference between the “nameplate” installed capacity and auxiliary power consumption used to operate the plant and its machinery) of coal plants in South Africa according to their location per supply area from 2021 to 2031. This indicates the decommissioning of six coal plants by the year 2031 namely Arnot, Camden, Grootvlei, Hendrina, Komati, and Kriel.

TABLE I. COAL PLANT CAPACITIES FROM 2021 – 2031

| Supply area | Coal plants | Nominal capacity (MW) in the respective years |      |      |      |      |      |      |
|-------------|-------------|-----------------------------------------------|------|------|------|------|------|------|
|             |             | 2021                                          | 2022 | 2025 | 2026 | 2027 | 2030 | 2031 |
| Mpumalanga  | Arnot       | 2100                                          | 2100 | 1860 | 1116 | 744  | 0    | 0    |
| Mpumalanga  | Camden      | 1481                                          | 1481 | 1111 | 741  | 370  | 0    | 0    |
| Mpumalanga  | Duvha       | 2875                                          | 2875 | 2875 | 2875 | 2875 | 2875 | 1917 |
| Mpumalanga  | Grootvlei   | 570                                           | 570  | 0    | 0    | 0    | 0    | 0    |
| Mpumalanga  | Hendrina    | 1140                                          | 1098 | 185  | 0    | 0    | 0    | 0    |
| Mpumalanga  | Kendal      | 3840                                          | 3840 | 3840 | 3840 | 3840 | 3840 | 3840 |
| Mpumalanga  | Komati      | 114                                           | 114  | 0    | 0    | 0    | 0    | 0    |
| Mpumalanga  | Kriel       | 2850                                          | 2640 | 2850 | 2375 | 1900 | 0    | 0    |
| Mpumalanga  | Kusile      | 2160                                          | 2880 | 4266 | 4266 | 4266 | 4266 | 4266 |
| Free State  | Lethabo     | 3558                                          | 3558 | 3558 | 3558 | 3558 | 3558 | 3558 |
| Mpumalanga  | Majuba      | 3843                                          | 3807 | 3843 | 3843 | 3843 | 3843 | 3843 |
| Limpopo     | Matimba     | 3690                                          | 3690 | 3690 | 3690 | 3690 | 3690 | 3690 |
| Mpumalanga  | Matla       | 3450                                          | 3450 | 3450 | 3450 | 3450 | 2875 | 1725 |
| Limpopo     | Medupi      | 4317                                          | 4317 | 4332 | 4332 | 4332 | 4332 | 4332 |
| Mpumalanga  | Tutuka      | 3510                                          | 3510 | 3510 | 3510 | 3510 | 3510 | 3510 |

### C. Renewable energy in South Africa

South Africa has huge untapped solar and wind energy potentials which can be harnessed to generate electricity. In

2021, the country had only 6105 MW of total installed capacity of renewable resources procured from Independent Power Producers (IPPs) of which 500 MW were from Concentrated Solar Power (CSP) plants, 2212 MW from solar photovoltaics, 3343 MW from onshore wind plants, and 50 MW from other resources. Fig. 2 shows these capacities alongside a 2031 case as projected in the Integrated Resource Plan (IRP) 2019, which estimates a total renewable energy (RE) mix of 26630 MW by 2031 [5].

Novel techniques and technology are being developed to reduce power delivery loss from inverter-driven generation. This includes optimal power injection theories. By increasing distribution efficiency, technical losses are reduced and in principle generation availability can be seen as increased [13]–[15].

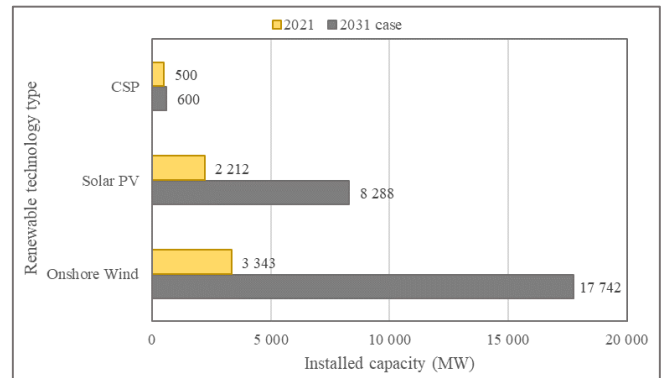


Fig. 2. South Africa's installed renewable capacities in 2021 and 2031 case

### D. Electricity demand in South Africa

In 2021, the peak national demand of South Africa was 35004 MW, and it is only set to increase in the future due to rapid increases in economic activity and income, changes in heating, fuel, transportation choices and rising urbanization. The Transmission Development Plan (TDP) 2022-2031 report forecasts a national system peak demand of 42000 MW under the moderate high scenario for the year 2031 [6]. Table II shows the peak demands for each of the supply areas in South Africa for the year 2021 and uses the TDP forecast for 2031.

TABLE II. PEAK SYSTEM DEMAND PER SUPPLY AREA FOR FISCAL YEAR 2021 AND 2031

| Supply area   | Peak demand (MW) |              |
|---------------|------------------|--------------|
|               | FY2021           | FY2031       |
| Eastern Cape  | 1584             | 1900         |
| Free State    | 1624             | 1948         |
| Gauteng       | 10614            | 12735        |
| Hydra Cluster | 205              | 246          |
| KwaZulu-Natal | 6051             | 7261         |
| Limpopo       | 2971             | 3565         |
| Mpumalanga    | 4223             | 5067         |
| North West    | 3280             | 3935         |
| Northern Cape | 685              | 822          |
| Western Cape  | 3768             | 4521         |
|               | <b>35004</b>     | <b>42000</b> |

## III. MODELLING IN FLEXTOL

### A. Overview

FlexTool is an open-source, node-based energy modelling tool where each node can represent either a region, area or country as seen in Fig. 3 [8].

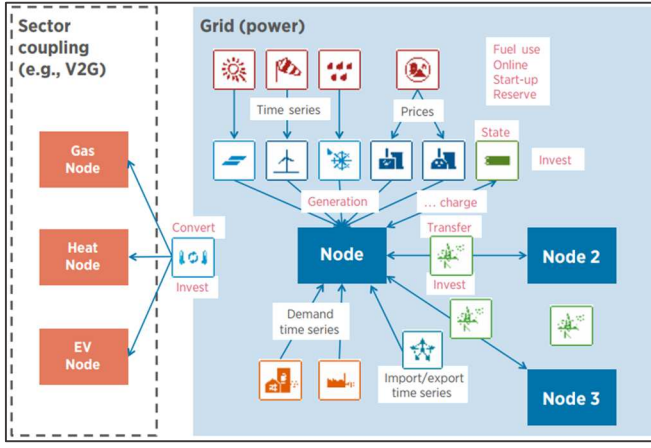


Fig. 3. Overview of energy modelling in FlexTool [8]

The nodes used for modelling and running the simulations in this paper represent South Africa's eleven supply areas as mentioned in the Generation Connection Capacity Assessment (GCCA) 2023 report [16], which included the nine provinces in South Africa, plus the Pelly, and Hydra clusters. However, Pelly was merged with Limpopo (one of the provinces) for this study, due to its proximity to Limpopo and lack of sufficient data to be regarded as a node. Thus, the South African power sector model used for this study comprises a ten-node system (represented by the ten supply areas), with each node having its separate variables such as generation, demand profile and node-to-node transfer connections to name a few.

#### B. Useful parameters

##### 1) Capacity factor (CF)

The capacity factor of a generating unit refers to the ratio of the electrical energy produced by this unit for a certain period of time, to the electrical energy that could have been produced at continuous full-power operation during the same period of time [17]. CF is calculated using (1).

$$CF = \frac{\text{Energy produced [MWh]}}{\text{Capacity [MW]} * t [\text{hours}]} \quad (1)$$

Where: *Capacity* (in MW) refers to the maximum amount of power that the generating unit can output onto the grid at full capacity and *t* is the time period in hours for which the CF is calculated, e.g., a month or a year.

As published in [18], the annual capacity factors per supply source in South Africa in 2021 indicated that nuclear plants had the highest capacity factor of 74.6%, followed by coal plants with 54.2%, CSP at 37.8%, wind plants at 35.8% and solar PV plants with a capacity factor of 26.4%.

##### 2) Surge Impedance Loading (SIL)

Surge Impedance Loading is defined as the power in which the reactive power of the transmission line becomes zero [21]. i.e., the reactive power generated by the shunt capacitance is consumed by the series inductance of the line. Equation (2) shows how SIL is calculated.

$$SIL = V_0^2 / Z_C \quad (2)$$

$$Z_C = \sqrt{L / C} \quad (3)$$

Where:  $V_0$  is the rated line voltage in volts (V),  $Z_C$  is the characteristic impedance of the line in ohms ( $\Omega$ ),  $L$  is the line inductance in henry (H) and  $C$  is line capacitance in farads (F).

#### IV. CASE STUDIES AND SIMULATION PROTOCOL

##### A. Case studies

The objective was to model different cases which might occur in a South African 2031 power sector scenario and analyze these results to better understand how the interconnected system might behave. Four cases studies were developed and in all the cases, the total renewable energy capacity that was added to the system by 2031 had been kept the same in line with IRP2019 [5]. However, the allocation of these capacities to the supply areas was altered based on either connection capacity constraints or transmission line expansions. Also, the grid was simplified using connectors greater than 110 kV between nodes (supply areas) and by applying the SIL conversion calculation to get the SIL ratings of the high-voltage transmission lines. The four cases studies that were developed are as follows:

##### 1) Case study I (CS I)

This case considered the grid connection capacity constraint as documented in the GCCA report [9], which states that the Northern Cape is highly constrained, and no new capacity can be added to this supply area without further grid strengthening. The allocation of renewable capacities in this case study was based on the rank of solar and wind capacity factors per supply area and the connection capacity available within that area. i.e., for solar PV allocation, Northern Cape has the best solar capacity factor of ~30.3%, but does not have any available connection capacity, and thus no new solar PV plant would be allocated to Northern Cape. Instead, the allocation criteria would move to the next best supply area for solar which is North-West with a solar capacity factor of ~30.1% and an available connection capacity of 3470 MW [16]. Thus, from the proposed 6076 MW of new solar capacity by 2031 [5], 3470 MW would be allocated to North-West. This process continued until all of the proposed solar capacity had been allocated and then it moved to allocate the proposed wind capacities with similar criteria. This case also does not include any battery storage capacity for the system. In summary, CS I was a case without transmission line expansion and no battery storage.

##### 2) Case study II (CS II)

This case considered the upstream strengthening of transmission lines to unlock more connection capacities in the greater Cape areas. The proposed transmission line expansions included in this study are:

- One 400kV line from Northern Cape to Hydra-cluster.
- One 765kV line from Northern Cape to North-West.
- One 765kV line from Northern Cape to Western Cape
- One 765kV line from Western Cape to Hydra cluster.
- One 765kV line from Eastern Cape to Hydra cluster.
- One 765kV line from Hydra cluster to Free state.
- One 765kV line from Free State to Gauteng.



The allocation criteria of renewable capacities in this case study were similar to that in CS I which was based on the rank of solar and wind capacity factors per supply area and the connection capacity available within that area. However, with these additional connection capacities, more renewable plants would be located in regions with high-capacity factors unlike the trade-offs in CS I due to the connection constraints. i.e., for solar PV allocation, Northern Cape would be allocated first due to its high solar capacity factor of ~30.3%, and availability of additional connection capacity. This process then continued until all of the proposed solar capacity had been allocated and then it moved to allocate the proposed wind capacities with similar criteria. This case also does not include any battery storage capacity for the system. In summary, CS II was a case with transmission line expansion but no battery storage.

### 3) Case study III (CS III)

This case had similar parameters to CS II in terms of RE allocation and transmission expansion. However, it also included battery storage in the system. Battery storage was allocated to regions that had installed solar PV plants. This study allocated a total of 2000 MW of battery storage capacity as proposed in IRP2019 [5] dividing it according to each supply area's installed solar capacity contribution. In summary, CS III was a case with transmission line expansion and 2000 MW of battery storage capacity.

### 4) Case study III-Alt (CS III-Alt)

This case was an alternate scenario of CS III in terms of its battery storage capacity. This case allocated a total of 3500 MW of battery storage capacity dividing it according to each supply area's installed solar capacity contribution. In summary, CS III-Alt was a case with transmission line expansion and 3500 MW of battery storage capacity.

Table III illustrates a summary of the dependencies and linkages between cases per category i.e., it shows that CS III and CS III-Alt models use the same parameters for RE allocation and transmission (Tx) expansion as developed in CS II but have different battery storage capacities.

TABLE III. SUMMARY OF CASE DEPENDENCIES PER CATEGORY

| Cases      | Category modelled in case # |              |                     |
|------------|-----------------------------|--------------|---------------------|
|            | RE allocation               | Tx expansion | Battery storage     |
| CS I       | CS I                        | ×            | ×                   |
| CS II      | CS II                       | CS II        | ×                   |
| CS III     | CS II                       | CS II        | CS III (2000MW)     |
| CS III-Alt | CS II                       | CS II        | CS III-Alt (3500MW) |

### B. Simulation protocol

The hourly generation profiles for each of the cases were obtained as one of the outcomes from the FlexTool simulation. For this study, the analysis of the hourly profiles was done for the worst week in the year of simulation (2031) for each of the cases. The worst week was the one which accounted for the most amount of load shedding.

The other outcomes from the FlexTool simulation that were analyzed in this paper included Variable Renewable Energy (VRE) share, VRE curtailment, load shedding and carbon dioxide (CO<sub>2</sub>) emissions. These were analyzed for each of the cases and discussed in the following chapter.

## V. RESULTS AND DISCUSSIONS

### A. Hourly profiles for the worst week

#### 1) Hourly profile for base case 2021

The worst week in the base case 2021 was week 23 (the first week of June) which accounted for roughly 38% of the total loss of load for the year. In the 2021 base case, we used the monthly availability factors for each of the coal power plants as reported in [19]. The maximum hourly loss of load for the worst week was 3413 MW as seen by the bright red spike in Fig. 4 at hour 3618, and most of it was caused due to lack of enough generation capacity (both from coal and RE plants) especially during the evening peak. It is also seen that a lot more of the peaking Open Cycle Gas Turbine (OCGT) plants are running during this week to compensate for the lack of generation capacity from coal and renewable plants. Week 23 will be the reference week for all the case studies to follow.

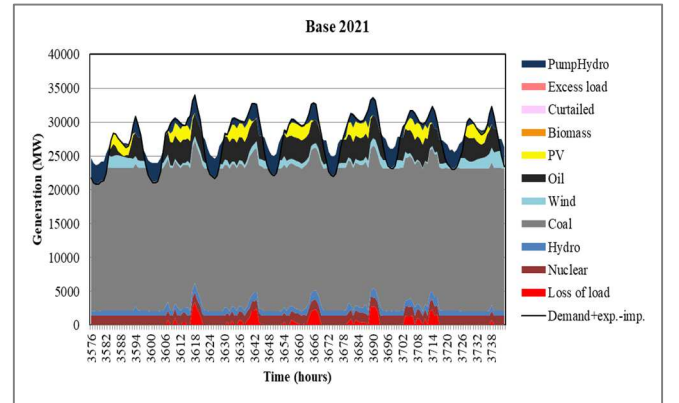


Fig. 4. Hourly profile of the worst week in 2021

#### 2) Hourly profile for CS I

Week 23 in CS I, accounted for roughly 23% of the total loss of load for the year. The maximum hourly loss of load for this week was 1441 MW as seen by the bright red spike in Fig. 5 at hour 3690. It is also seen that due to the increase in generation capacity in 2031, loss of load has significantly reduced. However, due to the lack of transmission expansion and lack of batteries, there is still some unmet demand especially during the evening peaks.

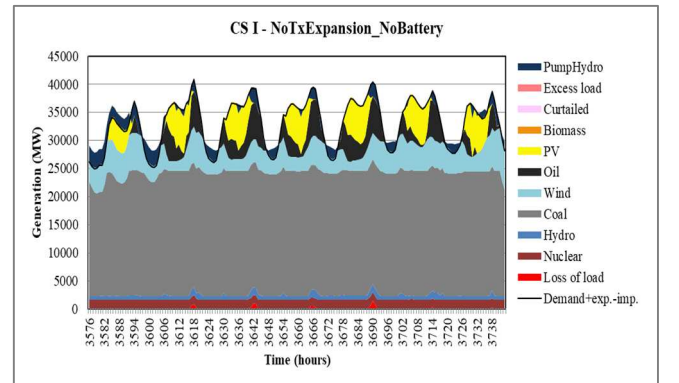


Fig. 5. Hourly profile of week 23 in CS I

#### 3) Hourly profile for CS II

Week 23 (the worst week) in CS II accounted for roughly 43% of the total loss of load for the year. The maximum hourly loss of load for this week was 2841 MW as seen by the bright red spike in Fig. 6 at hour 3618. Week 23 (the first week of June) was seen as the worst week having unmet

evening peak demand for most of the cases due to a couple of reasons. Firstly, June to August are winter months in South Africa, and thus, peak demand increases during evening hours due to the increased usage of heaters and geysers. Also, wind and solar capacity factors are at the lowest during these three winter months which add to a lower generation capacity than in summer months, especially without a battery storage option available as is the case for CS II.

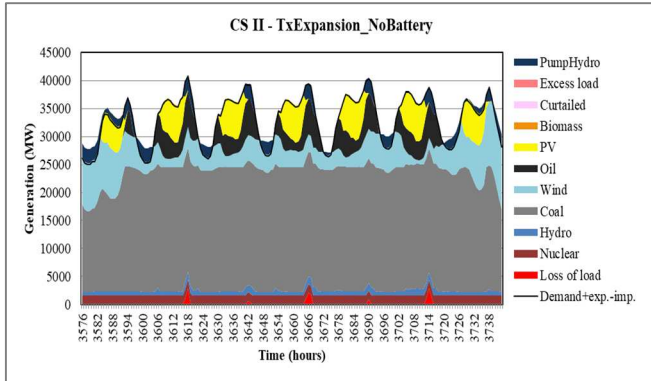


Fig. 6. Hourly profile of week23(worst week) in CS II

#### 4) Hourly profile for CS III-Alt

It can be seen from Fig. 7 that for CS III-Alt, there isn't any loss of load during its worst week (week 23). This is because of the addition of battery storage capacity, which can be charged up during the daylight hours and discharged during the evening peak to meet the high winter peak demands of South Africa.

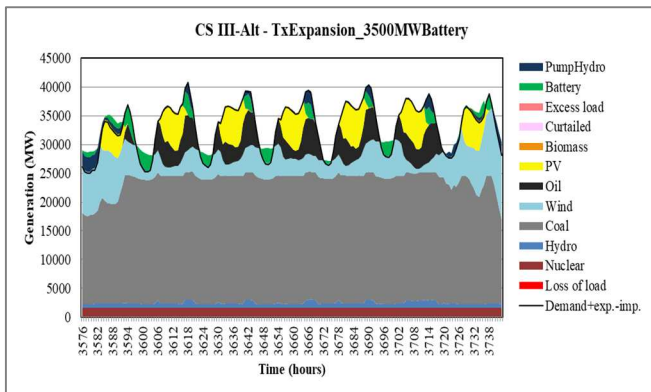


Fig. 7. Hourly profile of week23 in CS III-Alt

Having seen the hourly profiles for the 2031 cases, further comparisons based on various parameters can be conducted. These parameters include but are not limited to the following.

#### B. Variable Renewable Energy (VRE) share

The VRE share refers to the amount of renewable energy contributing to the electricity mix of that system for that particular year. The VRE share for this study takes into account all renewable energies including hydro. Fig. 8 shows that in 2021, the VRE share was just 12% and it is expected to increase to around 30% by 2031. Cases I to III-Alt, all have the same amount of installed RE capacity for 2031, however, Case I differs slightly (less) from the others due to its location of RE plants in lower yield supply areas due to the connection capacity constraint placed on CS I.

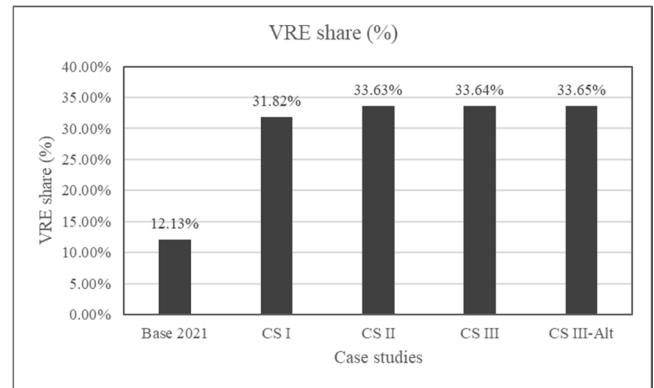


Fig. 8. Comparison of VRE share per case study

#### C. VRE curtailment

VRE curtailment refers to the deliberate reduction of power generation from renewable plants to balance energy demand and supply [20]. Curtailment can also occur due to a transmission constraint. It can be seen from Fig. 9 that base case 2021 and CS I did not have any VRE curtailment. It is also evident that curtailment is significantly reduced with the addition of battery storage as seen from comparing CS II to CS III and CSIII-Alt.

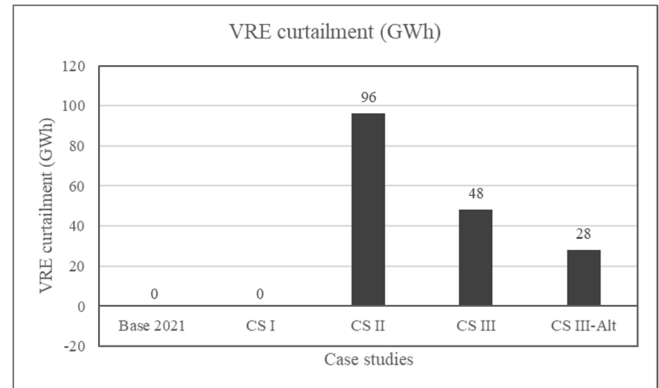


Fig. 9. Comparison of VRE Curtailment per case study

#### D. Load shedding

Load shedding refers to the systematic reduction of system load in order to balance the demand and supply of the system. Whereas loss of load is the unserved/unmet demand of the system. Fig. 10 indicates the comparison of loss of load due to load shedding in all four of the case studies.

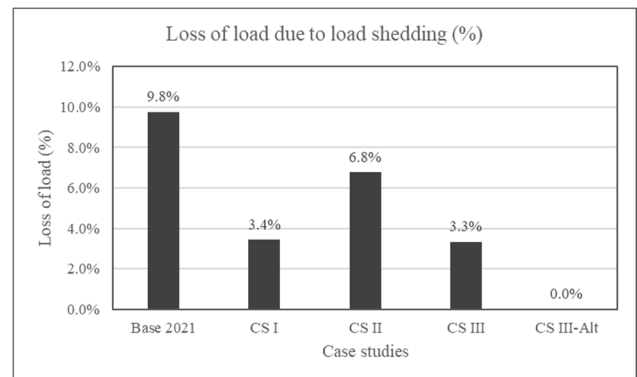


Fig. 10. Comparison of loss of load due to load shedding per case study

#### E. Carbon dioxide emissions

Information about the carbon dioxide emissions of a system is crucial in highlighting how clean a country's

generation mix is. Fig. 11 indicates a declining trend of CO<sub>2</sub> emissions with an increase in the renewable energy share for future 2031 scenarios.

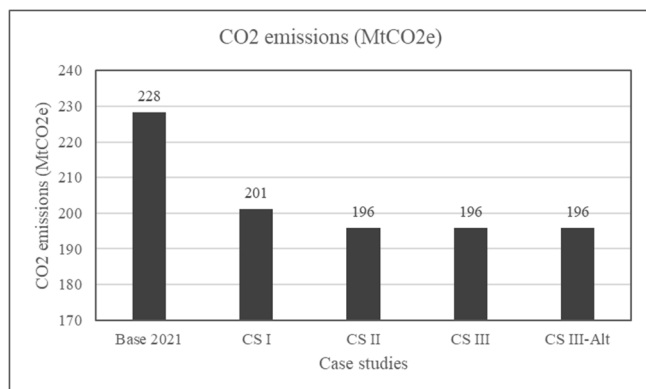


Fig. 11. Comparison of CO<sub>2</sub> emissions per case study

## VI. CONCLUSION

In this paper, the authors proposed and analyzed the 2031 South African power system energy mix using energy systems modelling in FlexTool. Simulation of the case studies considered various combinations of RE plant allocations, transmission line expansions and battery storage capabilities which would assist future grid planners to better understand how the load dynamics can be met with a generation mix biased towards renewable energy. The results indicated a higher share of renewable energy in the energy mix for 2031, reduced loss of load due to load shedding (from 6.8% in CS II to no loss of load in CS III-Alt) and reduced carbon dioxide emissions from the power sector from all the cases as compared to the 2021 base case. It also highlighted the importance of battery storage capacities as seen from CS III and CS III-Alt in minimizing VRE curtailment and unlocking additional capacity to meet the country's demand, especially during the evening peaks, which was also aided by strategic RE allocation and transmission expansion to further unlock additional grid capacity especially in the greater Cape areas. Cost-benefit analysis was excluded from this paper, but it remains the subject of ongoing research, and a further work incorporates conducting a power systems analysis in DigSILENT to test for grid stability of the system.

## ACKNOWLEDGMENT

The authors are grateful to the Department of Electrical Engineering, University of Cape Town and Energy Systems Research Group for proving the funds and infrastructure to carry out this research.

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